

AI Machine Learning Approach for Predicting the Habitability Potential of Exoplanets

1. Project Information

Project Name: ExoHabitAI

Submitted By: Gopika S

Submission Date: 19 February 2026

Deployment Platform: Streamlit Community Cloud

2. Project Abstract

The discovery of exoplanets has significantly increased in recent years. However, identifying which of these planets can potentially support life remains a major scientific challenge. Due to massive datasets and complex planetary parameters, traditional astronomical methods face limitations.

ExoHabitAI is a Machine Learning-based web application developed to predict the habitability potential of exoplanets using physical, orbital, and stellar characteristics. The system evaluates features such as planetary radius, equilibrium temperature, star temperature, and stellar radius to generate a habitability score and classification.

The application is fully deployed using Streamlit and publicly accessible through a web URL.

3. Problem Statement

With thousands of exoplanets discovered, manually identifying habitable candidates is inefficient. Astronomers require intelligent systems that can:

- Analyze multiple planetary parameters
- Predict habitability potential
- Rank planets based on likelihood of supporting life
- Reduce time spent on observational filtering

ExoHabitAI addresses this problem using supervised machine learning techniques.

4. Objectives

- Develop a machine learning model for habitability prediction
- Perform feature engineering and dataset preprocessing
- Train and evaluate classification models
- Deploy a publicly accessible web application
- Provide visual habitability insights

- Ensure full documentation and deployment readiness
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5. System Architecture

Application Flow

User → Streamlit Web Interface → ML Model → Prediction Output → Display Results

Explanation

1. User enters planetary and stellar parameters in the Streamlit UI.
 2. Input data is preprocessed inside the application.
 3. The trained ML model predicts habitability score.
 4. Result is displayed as:
 - Habitability Score
 - Classification (Habitable / Non-Habitable)
 - Visual indicators
-

6. Dataset & Feature Engineering

Dataset Source

- NASA Exoplanet Archive
- Public datasets (CSV format)

Features Used

- Planet Radius
- Planet Mass
- Equilibrium Temperature
- Orbital Period
- Star Effective Temperature
- Star Radius
- Stellar Luminosity

Data Processing Steps

- Handling missing values
 - Feature normalization
 - Feature selection
 - Target variable creation (Habitability Class)
-

7. Machine Learning Model

Model Used

- Random Forest Classifier
- XGBoost Classifier

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

The final trained model was saved as:

model.pkl

Stored inside:

models/

8. Deployment Details

Deployment Platform

Streamlit Community Cloud

Live Application URL

<https://exohabital-ai.streamlit.app/>

GitHub Repository

<https://github.com/GKSJ19/ExoHabitAI/tree/Gopikasivakumar>

9. Project Structure

ExoHabitAI/

```
|  
├── app.py  
├── backend/  
├── dashboard/  
├── models/  
│   └── model.pkl  
├── notebooks/  
├── requirements.txt  
└── LICENSE
```

10. requirements.txt

streamlit
numpy
pandas
scikit-learn
xgboost
joblib
plotly
matplotlib
seaborn

11. Testing & Validation

The deployed application was tested successfully for:

- Web app loads without errors
- User inputs processed correctly
- Prediction generated instantly
- Habitability score displayed
- Classification result displayed
- No runtime or dependency errors

Screenshots Included:

- Home Interface
 - Input Form
 - Prediction Output (HABITABLE Result)
 - Streamlit Deployment Dashboard
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12. License

This project is licensed under the MIT License.

The repository includes a LICENSE file as required.

13. Evaluation Milestone Completion

Milestone 1

- ✓ Data collection
- ✓ Feature engineering
- ✓ Dataset preparation

Milestone 2

- ✓ Model training
- ✓ Model evaluation
- ✓ Habitability ranking

Milestone 3

- ✓ ML integration
- ✓ Application UI development

Milestone 4

- ✓ Visualization dashboard
- ✓ Public deployment
- ✓ Full documentation

14. Public Access Confirmation

- GitHub repository is Public
- MIT License included
- Model file committed
- Application fully deployed
- Public Streamlit URL accessible
- All modules completed and tested

15. Final Declaration

I hereby confirm that the ExoHabitAI application has been successfully developed, tested, and deployed. The system is publicly accessible, fully functional, and ready for evaluation and demonstration.

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Date: 19 February 2026

ExoHabitAI - Streamlit

exohabitai-ai.streamlit.app

288.0

Star Effective Temperature (K)

5778.0

Star Radius (Solar Radius)

1.000

 Predict Habitability

 Prediction Complete!

Habitability Score

0.999

↑ 95%

 **HABITABLE**

Strong potential for life!

Activate Windows
Go to Settings to activate Windows.
[Manage app](#)

ExoHabitAI - Streamlit

exohabitai-ai.streamlit.app

210.0

Star Effective Temperature (K)

5772.0

Star Radius (Solar Radius)

1.000

 Predict Habitability

 Prediction Complete!

Habitability Score

0.566

↑ 57%

 **NOT HABITABLE**

Extreme conditions prevent life!

Activate Windows
Go to Settings to activate Windows.
[Manage app](#)